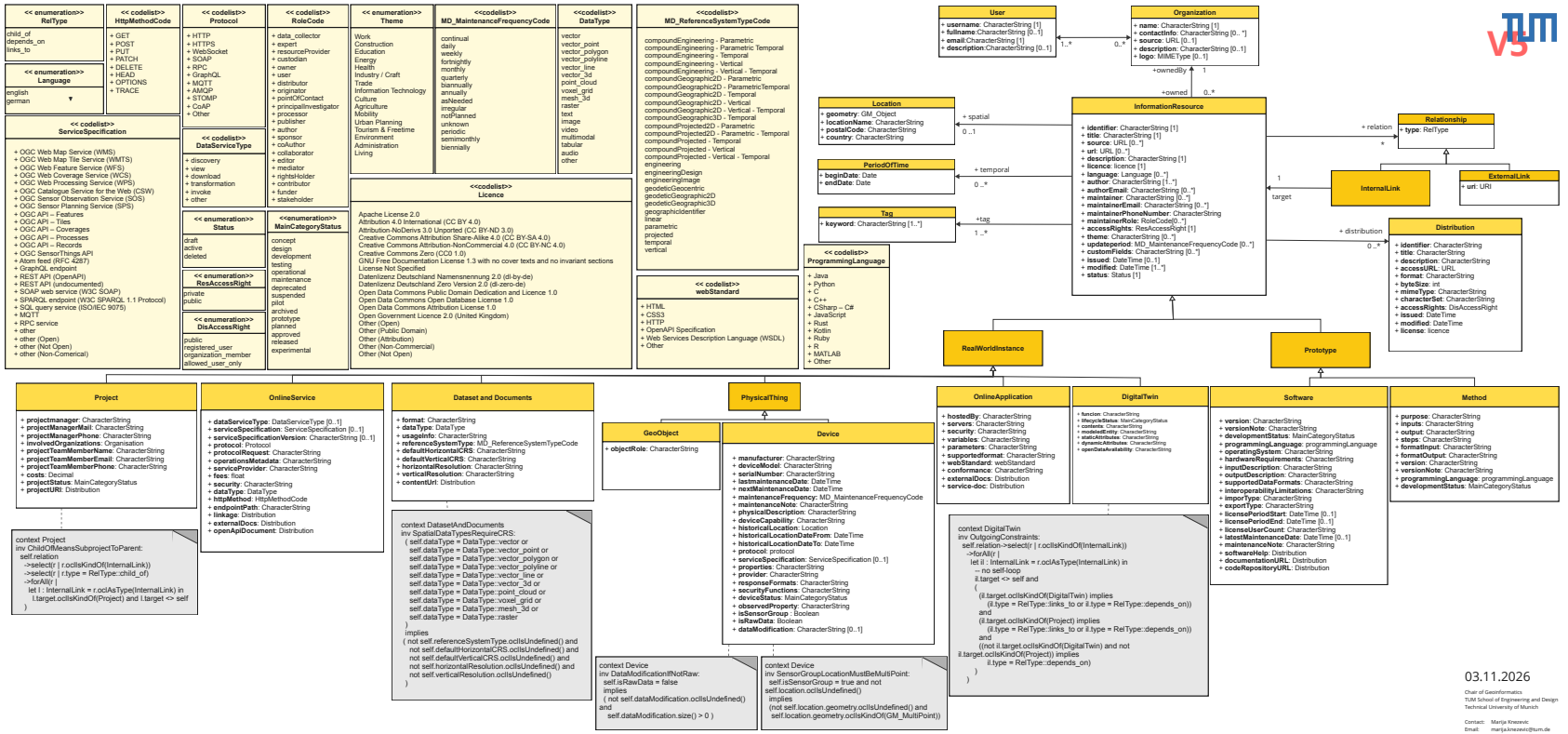

« enumeration »
RelType
child_of
depends_on
links_to
« codelist »
HttpMethodCode
+ GET
+ POST
+ PUT
+ PATCH
+ DELETE
+ HEAD
+ OPTIONS
+ TRACE
« enumeration »
Language
english
german
« codelist »
ServiceSpecification
+ OGC Web Map Service (WMS)
+ OGC Web Map Tile Service (WMTS)
+ OGC Web Feature Service (WFS)
+ OGC Web Coverage Service (WCS)
+ OGC Web Processing Service (WPS)
+ OGC Catalogue Service for the Web (CSW)
+ OGC Sensor Observation Service (SOS)
+ OGC Sensor Planning Service (SPS)
+ OGC API – Features
+ OGC API – Tiles
+ OGC API – Coverages
+ OGC API – Processes
+ OGC API – Records
+ OGC SensorThings API
+ Atom feed (RFC 4287)
+ GraphQL endpoint
+ REST API (OpenAPI)
+ REST API (undocumented)
+ SOAP web service (W3C SOAP)
+ SPARQL endpoint (W3C SPARQL 1.1 Protocol)
+ SQL query service (ISO/IEC 9075)
+ MQTT
+ RPC service
+ other
+ other (Open)
+ other (Not Open)
+ other (Non-Comerical)
« codelist »
Protocol
+ HTTP
+ HTTPS
+ WebSocket
+ SOAP
+ RPC
+ GraphQL
+ MQTT
+ AMQP
+ STOMP
+ CoAP
+ Other
« codelist »
DataServiceType
+ discovery
+ view
+ download
+ transformation
+ invoke
+ other
« enumeration »
Status
draft
active
deleted
« enumeration »
ResAccessRight
private
public
« enumeration »
DisAccessRight
public
registered_user
organization_member
allowed_user_only
« codelist »
RoleCode
+ data_collector
+ expert
+ resourceProvider
+ custodian
+ owner
+ user
+ distributor
+ originator
+ pointOfContact
+ principalInvestigator
+ processor
+ publisher
+ author
+ sponsor
+ coAuthor
+ collaborator
+ editor
+ mediator
+ rightsHolder
+ contributor
+ funder
+ stakeholder
« enumeration »
MainCategoryStatus
concept
design
development
testing
operational
maintenance
deprecated
suspended
pilot
archived
prototype
planned
approved
released
experimental
« enumeration »
Theme
Work
Construction
Education
Energy
Health
Industry / Craft
Trade
Information Technology
Culture
Agriculture
Mobility
Urban Planning
Tourism & Freetime
Environment
Administration
Living
« codelist »
MD_MaintenanceFrequencyCode
continual
daily
weekly
fortnightly
monthly
quarterly
biannually
annually
asNeeded
irregular
notPlanned
unknown
periodic
semimonthly
biennially
« codelist »
DataType
vector
vector_point
vector_polygon
vector_polyline
vector_line
vector_3d
point_cloud
voxel_grid
mesh_3d
raster
text
image
video
multimodal
tabular
audio
other
« codelist »
Licence
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Other (Open)
Other (Public Domain)
Other (Attribution)
Other (Non-Commercial)
Other (Not Open)
« codelist »
MD_ReferenceSystemTypeCode
compoundEngineering - Parametric
compoundEngineering - Parametric Temporal
compoundEngineering - Temporal
compoundEngineering - Vertical
compoundEngineering - Vertical - Temporal
compoundGeographic2D - Parametric
compoundGeographic2D - ParametricTemporal
compoundGeographic2D - Temporal
compoundGeographic2D - Vertical
compoundGeographic2D - Vertical - Temporal
compoundGeographic3D - Temporal
compoundProjected2D - Parametric
compoundProjected2D - Parametric - Temporal
compoundProjected - Temporal
compoundProjected - Vertical
compoundProjected - Vertical - Temporal
engineering
engineeringDesign
engineeringImage
geodeticGeocentric
geodeticGeographic2D
geodeticGeographic3D
geographicIdentifier
linear
parametric
projected
temporal
vertical
« codelist »
webStandard
+ HTML
+ CSS3
+ HTTP
+ OpenAPI Specification
+ Web Services Description Language (WSDL)
+ Other
« codelist »
ProgrammingLanguage
+ Java
+ Python
+ C
+ C++
+ CSharp – C#
+ JavaScript
+ Rust
+ Kotlin
+ Ruby
+ R
+ MATLAB
+ Other
User
+ username: CharacterString [1]
+ fullname:CharacterString [0..1]
+ email:CharacterString [1]
+ description:CharacterString [0..1]
1..*
0..*
Organization
+ name: CharacterString [1]
+ contactInfo: CharacterString [0..*]
+ source: URL [0..1]
+ description: CharacterString [0..1]
+ logo: MIMEType [0..1]
+ownedBy
1
+owned
0..*
Location
+ geometry: GM_Object
+ locationName: CharacterString
+ postalCode: CharacterString
+ country: CharacterString
+ spatial
0..1
PeriodOfTime
+ beginDate: Date
+ endDate: Date
+ temporal
0..*
Tag
+ keyword: CharacterString [1..*]
+tag
1..*
InformationResource
+ identifier: CharacterString [1]
+ title: CharacterString [1]
+ source: URL [0..*]
+ url: URL [0..*]
+ description: CharacterString [1]
+ licence: licence [1]
+ language: Language [0..*]
+ author: CharacterString [1..*]
+ authorEmail: CharacterString [0..*]
+ maintainer: CharacterString [0..*]
+ maintainerEmail: CharacterString [0..*]
+ maintainerPhoneNumber: CharacterString
+ maintainerRole: RoleCode[0..*]
+ accessRights: ResAccessRight [1]
+ theme: CharacterString [0..*]
+ updateperiod: MD_MaintenanceFrequencyCode [0..*]
+ customFields: CharacterString [0..*]
+ issued: DateTime [0..1]
+ modified: DateTime [1..*]
+ status: Status [1]
+ relation
*
Relationship
+ type: RelType
1
target
InternalLink
ExternalLink
+ url: URI
+ distribution
0..*
Distribution
+ identifier: CharacterString
+ title: CharacterString
+ description: CharacterString
+ accessURL: URL
+ format: CharacterString
+ byteSize: int
+ mimeType: CharacterString
+ characterSet: CharacterString
+ accessRights: DisAccessRight
+ issued: DateTime
+ modified: DateTime
+ license: licence
RealWorldInstance
Prototype
Project
+ projectmanager: CharacterString
+ projectManagerMail: CharacterString
+ projectManagerPhone: CharacterString
+ involvedOrganizations: Organisation
+ projectTeamMemberName: CharacterString
+ projectTeamMemberEmail: CharacterString
+ projectTeamMemberPhone: CharacterString
+ costs: Decimal
+ projectStatus: MainCategoryStatus
+ projectURI: Distribution
context Project
inv ChildOfMeansSubprojectToParent:
self.relation
->selectN(r.oclIsKindOf(InternalLink))
->select(r | r.type = RelType::child_of)
->forAll(r |
let i : InternalLink = r.oclAsType(InternalLink) in
i.target.oclIsKindOf(Project) and i.target <> self
)
OnlineService
+ dataServiceType: DataServiceType [0..1]
+ serviceSpecification: ServiceSpecification [0..1]
+ serviceSpecificationVersion: CharacterString [0..1]
+ protocol: Protocol
+ protocolRequest: CharacterString
+ operationsMetadata: CharacterString
+ serviceProvider: CharacterString
+ fees: float
+ security: CharacterString
+ dataType: DataType
+ httpMethod: HttpMethodCode
+ endpointPath: CharacterString
+ linkage: Distribution
+ externalDocs: Distribution
+ openApiDocument: Distribution
Dataset and Documents
+ format: CharacterString
+ dataType: DataType
+ usageInfo: CharacterString
+ referenceSystemType: MD_ReferenceSystemTypeCode
+ defaultHorizontalCRS: CharacterString
+ defaultVerticalCRS: CharacterString
+ horizontalResolution: CharacterString
+ verticalResolution: CharacterString
+ contentUrl: Distribution
context DatasetAndDocuments
inv SpatialDataTypesRequireCRS:
(self.dataType = DataType::vector or
self.dataType = DataType::vector_point or
self.dataType = DataType::vector_polygon or
self.dataType = DataType::vector_polyline or
self.dataType = DataType::vector_line or
self.dataType = DataType::vector_3d or
self.dataType = DataType::point_cloud or
self.dataType = DataType::voxel_grid or
self.dataType = DataType::mesh_3d or
self.dataType = DataType::raster
)
implies
(not self.referenceSystemType.oclIsUndefined() and
not self.defaultHorizontalCRS.oclIsUndefined() and
not self.defaultVerticalCRS.oclIsUndefined() and
not self.horizontalResolution.oclIsUndefined() and
not self.verticalResolution.oclIsUndefined()
)
PhysicalThing
GeoObject
+ objectRole: CharacterString
Device
+ manufacturer: CharacterString
+ deviceModel: CharacterString
+ serialNumber: CharacterString
+ lastmaintenanceDate: DateTime
+ nextMaintenanceDate: DateTime
+ maintenanceFrequency: MD_MaintenanceFrequencyCode
+ maintenanceNote: CharacterString
+ physicalDescription: CharacterString
+ deviceCapability: CharacterString
+ historicalLocation: Location
+ historicalLocationDateFrom: DateTime
+ historicalLocationDateTo: DateTime
+ protocol: protocol
+ serviceSpecification: ServiceSpecification [0..1]
+ properties: CharacterString
+ provider: CharacterString
+ responseFormats: CharacterString
+ securityFunctions: CharacterString
+ deviceStatus: MainCategoryStatus
+ observedProperty: CharacterString
+ isSensorGroup : Boolean
+ isRawData: Boolean
+ dataModification: CharacterString [0..1]
context Device
inv DataModificationIfNotRaw:
self.isRawData = false
implies
(not self.dataModification.oclIsUndefined()
and
self.dataModification.size() > 0)
context Device
inv SensorGroupLocationMustBeMultiPoint:
self.isSensorGroup = true and not
self.location.oclIsUndefined()
implies
(not self.location.geometry.oclIsUndefined() and
self.location.geometry.oclIsKindOf(GM_MultiPoint))
OnlineApplication
+ hostedBy: CharacterString
+ servers: CharacterString
+ security: CharacterString
+ variables: CharacterString
+ parameters: CharacterString
+ supportedFormat: CharacterString
+ webStandard: webStandard
+ conformance: CharacterString
+ externalDocs: Distribution
+ service-doc: Distribution
DigitalTwin
+ function: CharacterString
+ lifecycleStatus: MainCategoryStatus
+ contents: CharacterString
+ modelledEntity: CharacterString
+ staticAttributes: CharacterString
+ dynamicAttributes: CharacterString
+ operDataAvailability: CharacterString
context DigitalTwin
inv OutgoingConstraints:
self.relation->select(r | r.oclIsKindOf(InternalLink))
->forAll(r |
let ii : InternalLink = r.oclAsType(InternalLink) in
-- no self-loop
ii.target <> self and
(
(ii.target.oclIsKindOf(DigitalTwin) implies
(ii.type = RelType::links_to or ii.type = RelType::depends_on))
and
(ii.target.oclIsKindOf(Project) implies
(ii.type = RelType::links_to or ii.type = RelType::depends_on))
and
((not ii.target.oclIsKindOf(DigitalTwin)) and not
ii.target.oclIsKindOf(Project)) implies
ii.type = RelType::depends_on)
)
)
Software
+ version: CharacterString
+ versionNote: CharacterString
+ developmentStatus: MainCategoryStatus
+ programmingLanguage: programmingLanguage
+ operatingSystem: CharacterString
+ hardwareRequirements: CharacterString
+ inputDescription: CharacterString
+ outputDescription: CharacterString
+ supportedDataFormats: CharacterString
+ interoperabilityLimitations: CharacterString
+ importType: CharacterString
+ exportType: CharacterString
+ licensePeriodStart: DateTime [0..1]
+ licensePeriodEnd: DateTime [0..1]
+ licenseUserCount: CharacterString
+ latestMaintenanceDate: DateTime [0..1]
+ maintenanceNote: CharacterString
+ softwareHelp: Distribution
+ documentationURL: Distribution
+ codeRepositoryURL: Distribution
Method
+ purpose: CharacterString
+ inputs: CharacterString
+ output: CharacterString
+ steps: CharacterString
+ formatInput: CharacterString
+ formatOutput: CharacterString
+ version: CharacterString
+ versionNote: CharacterString
+ programmingLanguage: programmingLanguage
+ developmentStatus: MainCategoryStatus
03.11.2026
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Technical University of Munich
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